

Question 1

A hospital needs to assign three doctors (D1, D2, and D3) to three work shifts: Morning, Afternoon, and Night. Each doctor must be assigned exactly one shift while satisfying the given scheduling constraints. The objective is to determine a valid schedule using the Backtracking Search Algorithm, a Constraint Satisfaction Problem (CSP) technique in Artificial Intelligence. The algorithm systematically assigns shifts, checks constraints, and backtracks whenever a constraint is violated until a valid solution is found. This problem demonstrates AI-based scheduling and resource allocation problems efficiently.

Question 2

A robot operates in a 5×5 grid environment and must move from the Start (S) position to the Goal (G). Obstacles block certain paths, and the robot can move only up, down, left, or right. Each movement has a cost of 1, and the Manhattan Distance heuristic is used to guide the search. The objective is to determine the optimal path and total cost by applying the Breadth-First Search (BFS) algorithm while showing the node expansion process and queue updates.

Question 3

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment with blocked paths, risky zones, and limited sensing capability. Every movement has a cost of 1, while entering risky zones adds an extra cost of 2. The robot must dynamically update its knowledge, minimize the total cost, and safely reach the survivor. This problem demonstrates intelligent agents, online search, rational decision-making, and path planning in uncertain environments.